Kamageshwaran . D
Stat 50

ICE #1

in the lamp example compute P(B|D).

~~P(B|D) = P(D|B) P(B)~~

$$P(B|D) = \frac{P(D|B)\,P(B)}{P(D|A)\,P(A) + P(D|B)\,P(B) + P(D|C)\,P(B)}$$

$$= \frac{(0.01)(0.35)}{(0.015)(0.35) + (0.01)(0.35) + (0.02)(0.3)}$$

$$= \frac{0.0035}{0.00525 + 0.0035 + 0.006} = \frac{0.0035}{0.01475}$$

$$= 0.237288136$$

$$= 0.237$$

$\therefore P(B|D) = 0.237$